

A NONUNITAL OBSTRUCTION TO A CALKIN-ALGEBRA QUESTION IN ARXIV:2403.04137

FA-BANACH.001 LANE 12

STATUS

This is a counterexample to the final non-unitized question in Section 7 of Motakis–Puglisi, *The compact operators on c_0 as a Calkin algebra*, arXiv:2403.04137. It does not answer the preceding question about the unitization of $\mathcal{K}(\ell_2)$.

SOURCE STATEMENT

On page 22, after asking whether the unitization of $\mathcal{K}(\ell_2)$ is a Calkin algebra, the source adds the following broader formulation.

in [10] that every finite-dimensional semi-simple complex algebra is a Calkin algebra.

Problem 3. Is the unitization of $\mathcal{K}(\ell_2)$ isomorphic as a Banach algebra to the Calkin algebra of some Banach space?

More generally, if X is a Banach space with a shrinking Schauder basis, is $\mathcal{K}(X)$ isomorphic to the Calkin algebra of some Banach space?

The packet concerns only the final sentence:

if X has a shrinking Schauder basis, is $\mathcal{K}(X)$ isomorphic to a Calkin algebra?

PROOF INTUITION

Calkin algebras are automatically unital: the coset of the identity operator is their identity. By contrast, $\mathcal{K}(X)$ has an identity only when X is finite-dimensional. Thus any infinite-dimensional X with a shrinking basis, for example $X = \ell_2$, violates the conclusion before any finer operator-ideal structure enters.

Theorem 1. *Let X be an infinite-dimensional Banach space. Then $\mathcal{K}(X)$ is not isomorphic as a Banach algebra to the Calkin algebra $\text{Cal}(Y) = \mathcal{L}(Y)/\mathcal{K}(Y)$ of any Banach space Y , except for the zero quotient case, which is also impossible because $\mathcal{K}(X) \neq 0$. Consequently, taking $X = \ell_2$ gives a counterexample to the final non-unitized question from arXiv:2403.04137.*

Proof. First, $\mathcal{K}(X)$ is nonzero for every nonzero Banach space X , since rank-one operators are compact.

Date: 2026-06-20.

Suppose that $\mathcal{L}(Y)/\mathcal{K}(Y)$ is nonzero. Then $I_Y + \mathcal{K}(Y)$ is a multiplicative identity for the quotient algebra. If a surjective algebra isomorphism

$$\Phi : \mathcal{K}(X) \longrightarrow \mathcal{L}(Y)/\mathcal{K}(Y)$$

existed, and $e = \Phi^{-1}(I_Y + \mathcal{K}(Y))$, then for every $A \in \mathcal{K}(X)$,

$$\Phi(eA) = \Phi(e)\Phi(A) = \Phi(A), \quad \Phi(Ae) = \Phi(A)\Phi(e) = \Phi(A).$$

Injectivity gives $eA = A = Ae$. Hence e is a two-sided identity for $\mathcal{K}(X)$.

We now show that $\mathcal{K}(X)$ has no identity when X is infinite dimensional. If $e \in \mathcal{K}(X)$ were an identity for $\mathcal{K}(X)$, then for every $x \in X$ and every $x^* \in X^*$, the rank-one operator $x^* \otimes x$ would satisfy

$$e(x^* \otimes x) = x^* \otimes x.$$

Choosing x^* with $x^*(x) \neq 0$ yields $ex = x$. Thus $e = I_X$. But I_X is compact if and only if the closed unit ball of X is compact, equivalently if and only if X is finite-dimensional. This contradicts the assumption that X is infinite-dimensional.

If Y is finite-dimensional, then $\mathcal{K}(Y) = \mathcal{L}(Y)$ and the Calkin algebra is the zero algebra, not isomorphic to nonzero $\mathcal{K}(X)$. Therefore no Y works. Since ℓ_2 is infinite-dimensional and has a shrinking Schauder basis, $X = \ell_2$ is the desired counterexample. $\square$

LIMITATIONS

The preceding source question asks whether the unitization of $\mathcal{K}(\ell_2)$ is a Calkin algebra. The theorem above does not address that unitized problem. It also does not address the likely intended unitized generalization for $\mathcal{K}(X)$ when X has a shrinking Schauder basis.

VERIFICATION AND NOVELTY CHECK

The proof is purely algebraic and uses only the standard facts that Calkin algebras are unital and that $\mathcal{K}(X)$ is unital exactly in finite dimension. Cheap run indexes were searched for arXiv id 2403.04137 and close phrases involving $\mathcal{K}(\ell_2)$, unitization, shrinking bases, nonunitality, and Calkin algebras. Web searches on 2026-06-20 for exact and near-exact source phrases did not find a later record of this literal obstruction. Novelty confidence is moderate-low because the obstruction is elementary and may reflect a wording slip rather than the authors' intended unitized problem.

HUMAN REVIEW RECOMMENDATION

Check whether the final sentence of the source should be read literally or as an implicit shorthand for the unitization of $\mathcal{K}(X)$. The packet is a valid counterexample only to the literal non-unitized formulation.

REFERENCES

- [1] P. Motakis and D. Puglisi, *The compact operators on c_0 as a Calkin algebra*, arXiv:2403.04137.